

Central or Peripheral Venoarterial Extracorporeal Membrane Oxygenation for Pediatric Sepsis: Outcomes Comparison in the Extracorporeal Life Support Organization Dataset, 2000–2021

OBJECTIVES: Small studies of extracorporeal membrane oxygenation (ECMO) support for children with refractory septic shock (RSS) suggest that high-flow (≥ 150 mL/kg/min) venoarterial ECMO and a central cannulation strategy may be associated with lower odds of mortality. We therefore aimed to examine a large, international dataset of venoarterial ECMO patients for pediatric sepsis to identify outcomes associated with flow and cannulation site.

DESIGN: Retrospective analysis of the Extracorporeal Life Support Organization (ELSO) database from January 1, 2000, to December 31, 2021.

SETTING: International pediatric ECMO centers.

PATIENTS: Patients 18 years old young or younger without congenital heart disease (CHD) cannulated to venoarterial ECMO primarily for a diagnosis of sepsis, septicemia, or septic shock.

INTERVENTIONS: None.

MEASUREMENTS AND MAIN RESULTS: Of 1242 pediatric patients undergoing venoarterial ECMO runs in the ELSO dataset, overall mortality was 55.6%. We used multivariable logistic regression analyses to evaluate explanatory factors associated with adjusted odds ratios (aORs) and 95% CI of mortality. In the regression analysis of data 4 hours after ECMO initiation, logarithm of the aOR, plotted against ECMO flow as a continuous variable, showed that higher flow was associated with lower aOR of mortality ($p = 0.03$). However, at 24 hours, we failed to find such a relationship. Finally, peripheral cannulation, as opposed to central cannulation, was independently associated with greater odds of mortality (odds ratio, 1.7 [95% CI, 1.1–2.6]).

CONCLUSIONS: In this 2000–2021 international cohort of venoarterial ECMO for non-CHD children with sepsis, we have found that higher ECMO flow at 4 hours after support initiation, and central- rather than peripheral-cannulation, were both independently associated with lower odds of mortality. Therefore, flow early in the ECMO run and cannula location are two important factors to consider in future research in pediatric patients requiring cannulation to venoarterial ECMO for RSS.

KEYWORDS: critical care; extracorporeal membrane oxygenation; factual databases; pediatric; septic shock

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Pediatric sepsis is an ongoing cause of mortality and morbidity throughout the world, and according to the 2020 Surviving Sepsis Guidelines venoarterial extracorporeal membrane oxygenation (ECMO) support is an option for life support in those with refractory septic shock (RSS) (1). Historically, small, retrospective studies from over 15 years ago showed that



RESEARCH IN CONTEXT

- In cases of refractory septic shock in pediatric patients, the question when it comes to considering venoarterial ECMO support is whether to use central or peripheral cannulation?
- On the one hand, there is the need to restore high oxygen delivery to tissues, as prior studies have suggested that improved outcomes are associated with higher flows. On the other hand, are there greater risks with central vs. peripheral venoarterial ECMO cannulation?
- Therefore, in this report, we sought to investigate the relationship between venoarterial ECMO flows and cannulation site and mortality in pediatric patients with sepsis using the ELSO dataset, 2000–2021.

central cannulation (via sternotomy with direct cannulation of the right atrium and aorta) was associated with superior outcomes in RSS compared with peripheral cannulation (2–4). Then, an international dataset, 2006–2014, showed that using venoarterial ECMO flow greater than 150 mL/kg/min, as opposed to standard flow, was associated with “almost twice the survival rate” (5). Furthermore, a review of 2011–2018 literature concluded that a central vascular approach to venoarterial ECMO cannulation was feasible, achieved higher median percentage maximum flow, and had potential “for good patient outcomes in selected patients” (6). However, three articles published 2020–2024 indicate that peripheral ECMO cannulation may be a more viable alternative to central cannulation (7–9). And, in this context, we have learned from a number of publications that central cannulation is associated with increased cannula site bleeding (10) and nosocomial infections (11), while peripheral cannulation is associated with increased risk of cerebrovascular stroke, related to cannulation and ligation of the carotid artery (12). Also, central cannulation in nonpost-cardiotomy patients is fraught with logistic challenges such as cardiothoracic surgeon availability for cannulation and the difficulties of managing a patient with an open chest, which should be weighed against the idea that peripheral vascular cannulation lacks these barriers and could be faster to implement (6, 13).

Taking all the above factors together, in cases of RSS, the question when it comes to venoarterial ECMO support is whether to use central or peripheral cannulation? On the one hand, there is the need to restore high oxygen delivery to tissues, as prior studies have suggested that improved outcomes are associated with higher flows. However, on the other hand, are there greater risks with central vs. peripheral venoarterial ECMO cannulation? Therefore, we sought to investigate the relationship between ECMO flows and cannulation site and mortality in pediatric patients with sepsis using the Extracorporeal Life Support Organization (ELSO) dataset. We hypothesized that higher flow rates would be associated with lower odds of mortality, independent of cannula location.

METHODS

Our retrospective ELSO database study (titled “Peripheral vs. Central ECMO in Pediatric Septic Shock”) was approved by the Vanderbilt University Medical Center Institutional Review Board (IRB No. 220441) on May 10, 2022. This study met the ethical standards of human experimentation according to our IRB and our research work was in full compliance with the Helsinki Declaration of 1975. The need for patient/parent consent was waived by the IRB.

In 2022, we acquired access to the ELSO dataset after submitting the data registry request form to the organization (ELSO, Ann Arbor, MI). ELSO created a dataset by selecting patients 0–18 years old with the *International Classification of Diseases*, 9th Edition (ICD-9) or *International Classification of Diseases*, 10th Edition (ICD-10) diagnostic codes for sepsis, septicemia, or septic shock requiring venoarterial ECMO. We requested the time frame was from January 1, 2000, to December 31, 2021. Then, from the original ELSO data, we excluded subjects if they had more than one ECMO run, were converted from peripheral to central cannulation and vice versa, were converted from venoarterial ECMO to another ECMO modality, and those with potentially incorrect blood flow data entry, determined by the ELSO hard upper limit on flows (> 1.5 L/min for 0–28 d old, and > 10 L/min for > 28 d old [14]). Last, since the ELSO database also provided the primary diagnosis for cannulation along with contributing or associated diagnoses, we narrowed the initial ELSO cohort (see above) by excluding subjects with ICD-9

and ICD-10 codes for congenital heart disease (CHD). These cases constituted our final cohort (Fig. 1).

Study Groups, Parameters, and Outcomes

We created three comparison groups based on blood flow rate, cannulation location, and timing of ECMO cannulation. We defined “high blood flow” as greater than or equal to 150 mL/kg/min and “standard blood flow” as less than 150 mL/kg/min. We compared the “high” and “standard” blood flow groups at 4 and 24 hours of ECMO support because these timings are the only two time-stamped data that ELSO provides for flow rate. For cannula location, we compared central and peripheral venoarterial ECMO cannulation. For timing, epochs were created for ECMO runs during the first 15 years (early) and the last 5 years (late) of the 2000–2021 dataset.

Regarding parameters and demographic variables, we included gender, weight, age, race, year of ECMO

run, and center identification code. The center that provided the greatest number of patients contributed only 2.4% of the initial, overall data. We decided that centers contributing greater than 2% of the total database volume were “high-volume” centers. To control for high-volume ECMO centers, we divided centers into those that provided greater than 2% of the data and those that provided less than or equal to 2% of the data. Clinical variables included pre-ECMO peak serum lactate concentration, pre-ECMO pH, and pre-ECMO cardiac arrest. In relation to etiology, since we did not have patient bacteriological culture data, we used the organism listed in the sepsis diagnosis code (Table S1, <http://links.lww.com/PCC/C590>). However, this information only covered 30% of the initial overall cohort. Other pre-ECMO data—which were used as surrogates for severity of illness—included need for support modalities such as cardiopulmonary bypass (CPB), inhaled nitric oxide (iNO), use of vasoactive medicines, mechanical cardiac support (e.g.,

ventricular assist device), and other extracorporeal life support (e.g., plasma-pheresis, renal replacement therapy [RRT] therapies).

The primary outcome was mortality. Secondary outcomes included morbidities such as hours on ECMO, length of stay (LOS), and post-cannulation serum lactate concentration (mmol/L). Other secondary outcomes were the complications of central vs. peripheral ECMO cannulation, including: pericardial tamponade, cannula site bleeding, cannula problems, and moderate and severe hemolysis; CNS diffuse ischemia on CT or MRI, or CNS infarction found on ultrasound, CT, or MRI; pneumothorax requiring treatment, pulmonary hemorrhage; and, serum creatinine 1.5–3.0 mg/dL, and requirement for RRT.

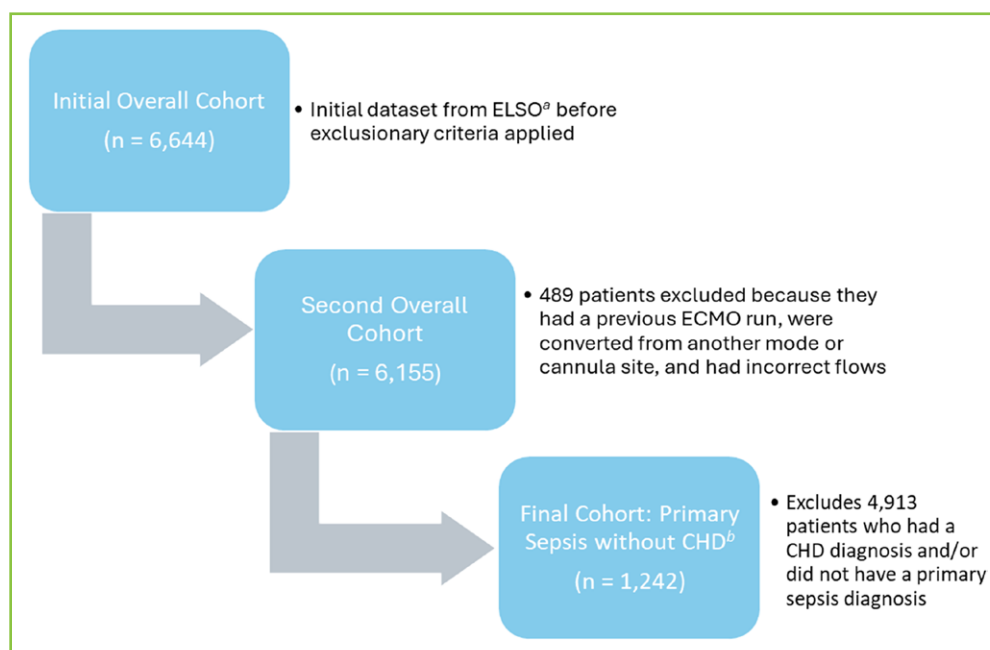


Figure 1. Flow chart depicting cohort population numbers with explanations of inclusion and exclusion criteria. The initial overall cohort was the raw dataset provided by Extracorporeal Life Support Organization (ELSO) that included patients 0–18 yr old with a primary or associated *International Classification of Diseases*, 9th Edition or *International Classification of Diseases*, 10th Edition diagnoses of sepsis, septicemia, or septic shock cannulated to venoarterial extracorporeal membrane oxygenation (ECMO). This dataset did not have any patients removed. The second overall cohort removed patients that had more than one ECMO run, cannula location conversion, ECMO modality conversion, and those with incorrect blood flow entry. This dataset included patients with congenital heart disease (CHD) and patients with a primary or associated diagnosis of sepsis. The final cohort only included patients with a primary diagnosis of sepsis, septic shock, or septicemia and without a diagnosis of CHD. ^aELSO. ^bCHD.

Statistical Analysis

We used the Wilcoxon rank-sum test for continuous variables and the chi-square test for categorical variables and performed multivariable logistic regression analysis to assess the association of explanatory factors of interest and mortality, presented as adjusted odds ratios (aORs) with a 95% CI. Explanatory factors included in the logistical regression analyses were as follows: patient age, year of ECMO, and center identification; pre-ECMO pH, cardiac arrest, and peak serum lactate concentration (mmol/L); use of iNO and vasoactive medicines; ECMO cannula location and ECMO blood flow at 4 and 24 hours; mediastinal cannula site bleeding and CNS infarction; serum creatinine 1.5–3 mg/dL and use of RRT; and mechanical cardiac support, CPB, and other extracorporeal life support. Logistic regression allowed for nonlinear model effects using regression splines. Missing values of explanatory factors included in the logistic regression were imputed using multiple imputation methods to make good use of partial information. Categorical variables were reported as frequencies (rates, proportions, or percentages) while continuous variables were reported with medians and interquartile ranges (IQRs). *p* values of less than 0.05 were considered significant.

RESULTS

Out of the 6644 patients in the initial 2000–2021 ELSO dataset, 1242 patients met our inclusion and exclusion criteria (Fig. 1). The mortality was 55.6% (691/1242) and the median age was 14.6 months (IQR, 0.4–90.0 mo) (**Table 1**). Pre-ECMO median serum lactate concentrations were 8 mmol/L (4.2–13.1 mmol/L) and a third of patients had a pre-ECMO cardiac arrest. Comparisons between the last 5 years and the first 15 years of the dataset are summarized in **Table S2** (<http://links.lww.com/PCC/C590>). Briefly, the data show that mortality—when known—in the later period vs. the earlier period was: (268/517 [51.8%] vs. 416/725 [57.4%]; difference, –5.6% [95% CI, –11.1% to 0.1%]; *p* = 0.09). That is, we failed to find an association between treatment in the later period and lower odds of mortality (odds ratio [OR], 0.8 [95% CI, 0.7–1.0]; *p* = 0.09). However, close inspection of the point estimate and the effect size show that we cannot exclude the possibility of up to one-fold greater odds of mortality in the latter period.

Flow and Cannula Location

The median ECMO pump flow of patients with a primary diagnosis of sepsis (and without CHD) was 101 mL/kg/min (IQR, 78–130 mL/kg/min) and 106 mL/kg/min (IQR, 79–135 mL/kg/min) at 4 and 24 hours, respectively. On univariate analysis of data at 4 hours, we failed to identify an association between high flow rate vs. standard flow and odds of mortality (OR, 1.3 [95% CI, 0.9–1.8]; *p* = 0.20) (**Table S3**, <http://links.lww.com/PCC/C590>). However, close inspection of the point estimate and effect size show that we are unable to exclude the possibility that high flow rate is associated with up to 1.8-fold greater odds of mortality. At 24 hours, high blood flow compared with standard flow was associated with greater odds of mortality (OR, 1.7 [95% CI, 1.2–2.4]; *p* = 0.0025).

There were 1034 patients (83.3%) cannulated peripherally, and 112 (9.0%) were cannulated centrally (**Table 1**). On univariate analysis, peripherally cannulated patients vs. central cannulation was associated with greater odds of mortality (OR, 1.6 [95% CI, 1.1–2.4]; *p* = 0.013) (**Table 2**). Furthermore, restricting the data to high blood flow rates, peripheral- vs. central-cannulation remained associated with greater odds of mortality (OR, 2.8 [95% CI, 1.3–6.1]; *p* = 0.007).

Comorbidities and Complications

Peripheral- vs. central-cannulation was associated with shorter median LOS (14 d [IQR, 3–37 d] vs. 26 d [IQR, 7–54 d]; *p* = 0.002) (**Table S4**, <http://links.lww.com/PCC/C590>). Peripheral- vs. central-cannulation was associated with lower odds of cannula site bleeding (OR, 0.2 [95% CI, 0.1–0.3]; *p* < 0.001), tamponade (OR, 0.2 [95% CI, 0.1–0.6]; *p* = 0.003), serum creatinine 1.5–3.0 mg/dL (OR, 0.6 [95% CI, 0.3–0.9]; *p* = 0.01), and greater odds of cannula problems (OR, 3.7 [95% CI, 1.4–10.3]; *p* = 0.002). Other comparisons are summarized in **Table S5** (<http://links.lww.com/PCC/C590>).

Multivariable Analysis

The logarithm of OR in the regression analysis of mortality was plotted as a trend against blood flow at 4 and 24 hours (**Fig. 2**). The graph of log OR of mortality shows an associated decrease with increased blood flow at 4 hours (**Fig. 2A**), while at 24 hours we failed

TABLE 1.**Demographic and Clinical Characteristics of All Patients Cannulated to Venoarterial Extracorporeal Membrane Oxygenation Primarily for Sepsis and Without Congenital Heart Disease**

Variable	All Patients (<i>n</i> = 1242)
Demographic variables	
Mortality, <i>n</i> (%)	691 (55.6)
Weight, kg, median (IQR)	10.3 (3.6–26.0)
Female, <i>n</i> (%)	554 (44.6)
Age, mo, median (IQR)	14.6 (0.4–90.0)
Age (groups) ^a , <i>n</i> (%)	
0–28 d	373 (30.0)
1 mo to 1 yr	219 (17.6)
1–5 yr	258 (20.8)
5–10 yr	145 (11.7)
10–18 yr	240 (19.3)
Race ^a , <i>n</i> (%)	
White	630 (50.7)
Black	204 (16.4)
Hispanic	147 (11.8)
Asian	97 (7.8)
Other	157 (12.6)
Epoch ^a , <i>n</i> (%)	
Early	725 (58.3)
Late	510 (41.0)
Center identification ^a , <i>n</i> (%)	
≥ 2%	96 (7.7)
< 2%	1139 (91.7)
Pre-ECMO variables	
Pre-ECMO peak serum lactate, mmol/L, median (IQR)	8.0 (4.2–13.1)
Pre-ECMO pH, median (IQR)	7.13 (7.0–7.24)
Pre-ECMO arrest, <i>n</i> (%)	406 (32.7)
ECMO variables	
Central cannulation, <i>n</i> (%)	112 (9.0)
Peripheral cannulation, <i>n</i> (%)	1034 (83.3)
Unknown cannulation location, <i>n</i> (%)	96 (7.7)
Pump flow at 4 hr, mL/kg/min, median (IQR)	101 (78–130)
Pump flow at 24 hr, mL/kg/min, median (IQR)	106 (79–135)
Hours on ECMO, median (IQR)	106 (48–177)
Cardiac indication, <i>n</i> (%)	527 (42.4)
Pulmonary indication, <i>n</i> (%)	526 (42.4)
Extracorporeal cardiopulmonary resuscitation, <i>n</i> (%)	182 (14.7)
Unreported indication, <i>n</i> (%)	7 (0.6)

ECMO = extracorporeal membrane oxygenation, IQR = interquartile range.

^aUnreported, *n* = 7 (0.6%).

TABLE 2.**Mortality Between Peripheral and Central Cannulation Overall and at High Blood Flow**

Outcome	Peripheral	Central	Risk Difference (95% CI)	OR (95% CI)	<i>p</i>
Overall, <i>n</i> (%)					
Died	576 (55.7)	49 (43.8)	11.9% (2.2–21.4%)	1.6 (1.1–2.4)	0.013
Survived	451 (43.6)	63 (56.3)			
Unreported	7 (0.7)	0 (0)			
High blood flow (≥ 150 mL/kg/min), <i>n</i> (%)					
Died	74 (64.3)	14 (38.9)	25.4% (6.8–42.7%)	2.8 (1.3–6.1)	0.007
Survived	41 (35.7)	22 (61.1)			
Unreported	0 (0)	0 (0)			

OR = odds ratio.

**AT THE BEDSIDE**

- In the Extracorporeal Life Support Organization (ELSO) 2000–2021 dataset we identified 1242 pediatric sepsis patients without congenital heart disease who underwent venoarterial extracorporeal membrane oxygenation (ECMO) runs, with an overall mortality of 55.6%.
- These data show that higher ECMO flow at 4 hours after initial support, and central- rather than peripheral-cannulation, were both independently associated with lower odds of mortality
- Therefore, ECMO flow early in the ECMO run and cannula location are two important factors to consider in future research in pediatric patients requiring cannulation to venoarterial ECMO for refractory septic shock.

to identify such an association. **Figure S1** (<http://links.lww.com/PCC/C590>) contains the graphs of nonlogarithmic aOR of mortality for both 4 and 24 hours. Even though we failed to find an association between increasing flows at 24 hours and the odds of mortality, the graph (**Fig. S1b**, <http://links.lww.com/PCC/C590>) of the nonlogarithmic OR of mortality demonstrates that we cannot exclude the possibility that flows above 150 mL/kg/min are associated with increased odds of mortality. **Figure S2** (<http://links.lww.com/PCC/C590>) contains the graphs of other continuous variables used in regression analyses.

Higher peak pre-ECMO serum lactate concentration, lower pre-ECMO pH, and younger age were each associated with greater odds of mortality, while we failed to show such an association with year of ECMO support. Categorical variables that were associated with greater aOR of mortality included: peripheral cannula location, vs. central (aOR, 1.7 [95% CI, 1.1–2.6]); pre-ECMO arrest, vs. not (aOR, 1.8 [95% CI, 1.3–2.3]); and CNS infarction, vs. not (aOR, 2.5 [95% CI, 1.5–4.0]) (**Table 3**).

DISCUSSION

In this retrospective study of the ELSO 2000–2021 database, we have examined the association between flow and cannula location with mortality among pediatric and neonatal non-CHD patients cannulated to venoarterial ECMO primarily for sepsis, septicemia, or septic shock. On univariate analysis the data showed that at 4 hours we were unable to exclude the possibility that high flow rate was associated with up to 1.8-fold greater odds of mortality. However, at 24 hours, the statistics were clearer (albeit in the same direction as the 4-hr findings), with high blood flow compared with standard flow being associated with 1.7-fold greater odds of mortality (95% CI, 1.2–2.4). In contrast, the multivariable regression analysis showed an opposite association with higher flow being independently associated with lower adjusted odds of mortality at the 4-hour time point. The regression analysis also revealed that peripheral rather than central cannulation was independently associated with greater odds of mortality.

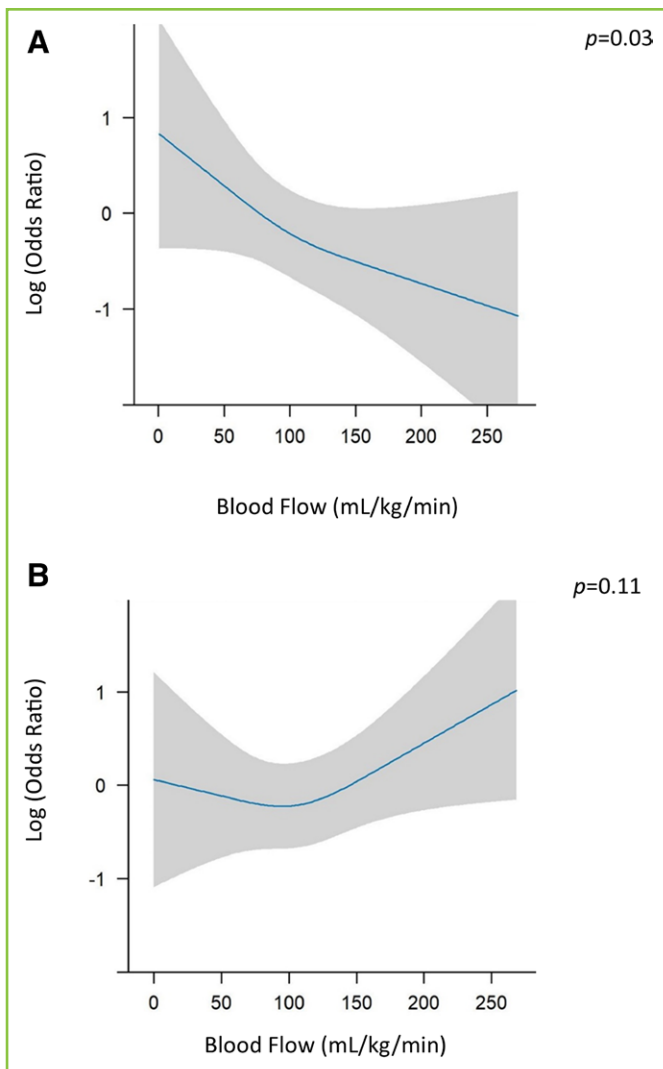


Figure 2. Regression curves plotting the relationship of flow to mortality at 4 and 24 hr continuously. Adjusted odds ratios for mortality are logarithmically converted on the Y-axis and extracorporeal membrane oxygenation blood flow in mL/kg/min are plotted on the X-axis to provide a continuous relationship of blood flow to mortality. **A**, Adjusted odds of mortality with increasing blood flow at 4 hr. **B**, Adjusted odds of mortality with increasing blood flow at 24 hr. Odds ratio of 1 means $\text{Log}(1) = 0$. Odds ratio of 0.5 means $\text{Log}(0.5) = -0.3$. Odds ratio of 2 means $\text{Log}(2) = +0.3$. $p < 0.05$ is considered significant.

Our retrospective study cannot determine decisions about optimal blood flow rate during venoarterial ECMO, but the multivariable analyses using 4-hour time point data showed an association between higher blood flow and lower adjusted odds of mortality in a rate-dependent manner. This relationship may be due to two reasons. This observation may reflect the benefits of better ECMO support, such as appropriate cannula positioning and venous return

pressures. Alternatively, children with sepsis requiring ECMO often present with shock that is associated with depressed left ventricular function and lower cardiac index (15, 16), requiring high inotropic or vasoactive medications (15, 17), and the benefits of early restoration of patient blood flow and perfusion with extracorporeal support are time-limited. For example, these patients—especially neonates—are known to have a stress cardiomyopathy related to RSS and may well respond to the initial high blood flow after cannulation (18, 19). Of note, the previous 2006–2014 venoarterial ECMO international dataset study showed that a flow greater than 150 mL/kg/min, as opposed to standard flow (even to 100 mL/kg/min), was associated with “almost twice the survival rate” (5). While our 4-hour data indicate that targeting higher flows may be reasonable, we recognize this is not always feasible. Even if the flows are less than 150 mL/kg/min, our data suggests that maximizing flows as much as possible early in the ECMO run may be associated with benefit. The graph (**Fig. S1a**, <http://links.lww.com/PCC/C590>) depicting the nonlogarithmic aOR of mortality at 4 hours demonstrates that flows greater than 100 mL/kg/min start to show an association with lower odds of mortality. Since we failed to identify a relationship between blood flow and odds of mortality at 24 hours, it is difficult to understand the apparent differences in associated odds of mortality between 4 and 24 hours and this warrants further study. Also, we need to better understand the apparent discrepancy between the univariate and the multivariate analyses.

The association between cannula location and odds of mortality in the regression analysis reiterate the findings in previous studies (2–4, 20). Central-cannulation in patients may result in higher flows earlier and meet the demands of oxygen delivery sooner. The median blood flow in patient runs with central-cannulation was higher than flow in patient runs with peripheral-cannulation in our dataset (**Table S6**, <http://links.lww.com/PCC/C590>), and in the literature (6, 20). In a 2002–2016 bi-National retrospective dataset, Schlapbach et al (20) reviewed 80 patients with septic shock on ECMO with a multivariable model and found that central cannulation was independently associated with lower odds of mortality. Even though flow was not an independent explanatory variable associated with mortality in this study, median flows in centrally cannulated patients were much higher than

TABLE 3.**Multivariable Logistical Regression Analysis of Categorical Risk Factors for Mortality in All Patients**

Variable	Adjusted OR (95% CI)	<i>p</i>
Peripheral location	1.7 (1.1–2.6)	0.02
Pre-extracorporeal membrane oxygenation arrest	1.8 (1.3–2.3)	< 0.001
CNS infarction	2.5 (1.5–4.0)	< 0.001
Center identification < 2 %	1.5 (0.9–2.4)	0.08
Mediastinal cannula bleeding	1.1 (0.3–4.2)	0.9
Creatinine 1.5–3.0 mg/dL	0.9 (0.6–1.4)	0.6
Renal replacement therapy	1.1 (0.9–1.5)	0.3
Vasoactive medications	1.2 (0.3–4.2)	0.3
Mechanical cardiac support	0.1 (0.01–1.8)	0.1
Cardiopulmonary bypass	6.5 (0.6–68.2)	0.1
Inhaled nitric oxide	0.8 (0.6–1.1)	0.2
Other extracorporeal support	1.4 (0.8–2.5)	0.2

OR = odds ratio.

in peripherally-cannulated patients (173 vs. 129 mL/kg/min, respectively [20]). Since early high flow was independently associated with lower odds of mortality in our study, the difference in 4-hour flow may explain the results of Schlapbach et al (20). Furthermore, our study shows central- vs. peripheral-cannulation was associated with fewer cannula problems, defined by ELSO as those events requiring intervention (14). This eventuality could potentially lead to fewer interruptions of flow and cardiac output.

Another factor that may influence outcomes within the centrally cannulated patients is medical center experience. Central-cannulation for septic shock is done at high-volume ECMO centers (4, 6) that are likely proficient at performing ECMO overall, and for RSS. We tried to control for high-volume ECMO centers by using the center identification variable in our multivariable model. Even though we failed to find an association between center identification and odds of mortality in the regression analysis, the point estimate and the effect size fail to exclude the possibility that management in centers with less than 2% volume was associated with up to 2.4-fold greater odds of mortality. Larger sample size is now needed in future research to address this question.

There are some limitations to our retrospective study of ELSO data. First, the study provides only a single time point, with episodic data for variables that

change over time such as blood flow rate, laboratory values, and hemodynamic parameters. Second, certain variables had missing or incorrect data, such as height (needed to calculate body surface area for cardiac index estimation), which limited the variables we could include in our work. Third, we did not have access to bacteriological culture data, which limited our ability to identify a sepsis-causative organism in many cases. Fourth, sepsis is a heterogeneous disease with changing physiology, even within the same patient, and separating these differences into phenotypes (21, 22) is not possible in the ELSO dataset, but it is something we should aspire to like the work with computational phenotypes in large datasets (23, 24). Last, there is a problem with the sample size that we had available: the number of patients with central-cannulation is much lower than (and unbalanced by) the number of patients undergoing peripheral-cannulation patients, which likely resulted in the variability in our analyses.

In conclusion, in our retrospective analysis of the 2000–2021 ELSO dataset of pediatric patients without CHD undergoing venoarterial ECMO for RSS, we have found that higher ECMO flow at 4 hours and use of central-cannulation were both independently associated with lower odds of mortality. Flow early in the ECMO run and cannula location are two important factors to consider in future research in pediatric

patients requiring cannulation to venoarterial ECMO for RSS.

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